Formal equivalence checking of PyTorch programs with ESBMC

BMC for fusion/rewrite correctness — and how the encoding compares to interactive proof

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Proof-of-concept · status update (expert version)

Problem & threat model

Optimising transformations — **operator fusion, kernel rewrites, compiler passes** — must be semantics-preserving. We decide:

$$\forall\, X, W \text{ with entries in } [-10, 10]. \quad P_{\text{unfused}}(X, W) \;\equiv\; P_{\text{fused}}(X, W)$$

- **Testing** under-approximates: it samples a finite input set.
- We want **exhaustive** equivalence over an input region, with **bit-precise IEEE-754** (where fusion breaks: reassociation, rounding).
- Two verdicts: **proof** (UNSAT) or a **concrete counterexample** (SAT).

Checked under a **sequential IEEE-754 semantics**; inputs are bounded, so **NaN/Inf are excluded** — equivalence is proven over finite FP domains. **Not modelled:** backend optimisations (FMA, cuBLAS), parallel/tree reductions, scheduling.

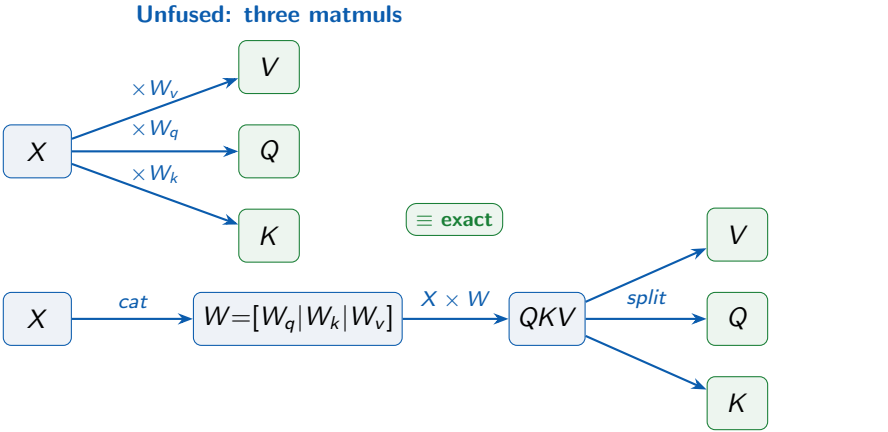
Running example: QKV projection

$$\text{unfused: } Q = XW_q,\ K = XW_k,\ V = XW_v$$

$$\text{fused: } W = [\, W_q \mid W_k \mid W_v \,], \quad QKV = XW, \quad \text{split}$$

The fused column j is the **identical multiply–add sequence** as the corresponding unfused projection $\Rightarrow$ the two are equal **bit-for-bit under the same evaluation order** (sequential IEEE-754), not merely within a tolerance. Both claims are checked.

The two programs, side by side



The embedding (1/2): encoding design

The program **acts as the specification** under the assumed operational model (correctness depends on the OM's faithfulness) — no separate spec to maintain.

- **Tensors** → nested lists; **ops** → operational model `torch.py` (`mm/matmul/cat/split/allclose`), pure-Python, float-typed, compiled into the ESBMC binary.
- **Inputs** → symbolic `nondet_float()` with `__ESBMC_assume` bounds (excludes NaN/Inf ⇒ finite FP domain).
- **Shapes** → dimensions are **fixed and well-typed**; shape compatibility is assumed.
- **Property** → exact (`==` / `allclose(rtol=0,atol=0)`) or tolerance (`allclose` defaults).
- **Two encodings**: scalar-unrolled (no OM) *and* torch-native.

The embedding (2/2): example harness

```
X   = [[bounded() for _ in range(D)] for _ in range(S)]    # symbolic
QKV = torch.mm(X, torch.cat([Wq, Wk, Wv], dim=1))         # fused weight (cat, dim=1)
assert torch.allclose(torch.mm(X, Wq), split_Q(QKV), 0.0, 0.0)
```

Symbolic bounded inputs; the matmul goes through the operational model; the property asserts exact (zero-tolerance) equivalence of the unfused and fused Q. *(In the PoC, `cat/split` are realised by manual column indexing — `torch.cat/split` are unwinding-heavy.)*

Pipeline & soundness

Python frontend → **GOTO** → symbolic execution → **VCs** → SMT.

- Fixed dims ⇒ loops fully unwound; **unwinding assertions ON** (no vacuous SUCCESS on truncation). Symbolic dims ⇒ `--k-induction` with convergence.
- **Bit-precise IEEE-754** FP theory (Bitwuzla / Z3); rounding modelled, not abstracted.
- **UNSAT** ⇒ **proved** over the region; **SAT** ⇒ **falsifying assignment**.
- TCB: Python frontend + operational model + SMT solver. **OM validated** vs real PyTorch (`validation/`): `cat`/`split`/`allclose` bit-exact, `mm` within ∼4e-13 (torch's reduction order differs).
- **Not modelled**: backend-specific optimisations (FMA, parallel/tree reductions, cuBLAS). **Intended use: reference-semantics equivalence**, not production-kernel equivalence.

Results

8 / 8 — QKV proved *exact* and *tolerance*, in *scalar* and *torch-native* encodings; each clean target paired with a refuted mutant (Bitwuzla, fixed dims).

Target	**predicate**	**verdict**	**time**
`qkv_equivalence_exact`	scalar ==	✓ SUCCESSFUL	6 s
`qkv_equivalence`	scalar tolerance	✓ SUCCESSFUL	12 s
`qkv_equivalence_torch_exact`	`allclose(0,0)`	✓ SUCCESSFUL	122 s
`qkv_equivalence_torch`	`allclose` def.	✓ SUCCESSFUL	124 s
`*_buggy` ($\times 3$)	refutation	✗ violated (c.ex.)	35–338 s
`bias_linear` (+ buggy)	$XW+b \equiv [X\|1][W;b]$	✓ / ✗	35–135 s

Legend: ✓ SUCCESSFUL = property holds (UNSAT); ✗ VIOLATED = counterexample found (SAT). The `*_buggy` targets are intentional mutants, so a counterexample is the *desired* outcome (non-vacuity check).

The embedding was *not* free

Driving real PyTorch-style code through ESBMC required fixing the embedding path — **4 contributions merged upstream**:

- `torch` **operational model** (#5120).
- **Nested-list value/type across returns & copies** (#5111, #5113; fixes #5102/#5103).
- **Homogeneous nested-list FP element-type** (#5131; fixes #5129) — found, root-caused, and fixed by this PoC.

The *embedding cost* here was partly tool-engineering — now amortised for everyone.

ESBMC (BMC) vs Lean (ITP) + LLM agent

Aspect	**ESBMC (BMC)**	**Lean (ITP) + LLM agent**
Embedding effort	program acts as spec under the OM (write both, assert $\equiv$)	embed tensor + FP semantics, **state** the theorem
Automation	push-button: auto VC-gen + SMT	interactive; agent drives tactic search, proofs need guidance
FP semantics	**bit-precise IEEE-754** native	modelled/axiomatised; bit-level reasoning heavy
Counterexamples	**concrete falsifying input** (SAT)	none by default
Scope	**bounded** (fixed sizes)	**unbounded / size-general**
TCB	frontend + OM + solver	small proof kernel

When to use which

- **Gate rewrites with BMC (ESBMC):** automatic, bit-precise, bug-finding, on concrete shapes — fast CI feedback, and a *counterexample* when it breaks.
- **Reach for ITP (Lean) for the size-general law:** $\forall$ shapes, machine-checked.
- They **compose**: BMC to find bugs and certify shipped configurations; ITP for the general theorem — where the **encoding + proof effort** and the **lack of FP counterexamples** is the real cost.

For "is this fusion correct for the shapes we deploy?", push-button BMC with bit-precise FP and counterexamples is highly effective in practice.

Empirical check: we encoded it in Lean too

Same QKV equivalence in **Lean 4** (agent = LLM-authored proof):

- The order-preserving fusion is a **definitional identity** ⇒ proved by `rfl` for **all dimensions**, over `Int` *and* `Float`.
- A **reassociating** fusion is sound over $\mathbb{Z}/\mathbb{R}$ (`omega`/`ring`) but **false in Float**; core Lean (opaque `Float`) can't prove it / gives **no counterexample**.
- **Verified with a real FP library:** *FloatSpec* proves the fusion over genuine **IEEE-754 rounding** ($\forall$ dims, `rfl`) — Lean *does* do bit-precise FP — but FloatSpec is a **noncomputable** proof model, so still **no concrete eval / counterexample**; ESBMC synthesises one.

	Lean	**ESBMC `--ir`**	**ESBMC `--floatbv`**
Order-preserving QKV	`rfl`, $\forall$ dims, $\sim$5 s	bounded, $\sim$3 s	bounded, 6–110 s
Reassociation / c.ex.	no c.ex.	FP-blind, real c.ex.	**FP counterexample**

ITP wins generality; `--ir` matches Lean's reals & speed; `--floatbv` **wins bit-precise FP + counterexamples**.

Verified in `lean/floatspec/`, `lean/Bits.lean`.

Related work: translation validation (Alive2)

The closest relative is **Alive2** (`alive2.llvm.org`) — translation validation for **LLVM IR** compiler optimisations. Same idea (prove-or-counterexample), one layer down.

	Alive2	**This work**
Target	LLVM IR optimisations	PyTorch fusions/rewrites
Engine	SMT (Z3), **refinement**	ESBMC → SMT, **equivalence**
Hard semantics	`undef`/`poison`/UB	**bit-precise IEEE-754** FP
Verdict	refines / counterexample	proved / counterexample
Scope	per-function (no inter-proc.)	bounded, fixed shapes

Different *hard problem*: Alive2's is UB/poison refinement; ours is floating-point (reassociation, rounding). Both are bounded/local — the size-general theorem is the ITP (Lean) end of the spectrum. **ITP side:** *TorchLean* (`github.com/nktkt/leanx`) formalises NNs in Lean 4 with IEEE-754 binary32 — Lean *can* do bit-precise FP, with a dedicated library.

Which tool for *this* task? (original ≡ optimised PyTorch)

Recommendation: ESBMC (BMC), default `--floatbv`. It matches the task:

- ✓ **Checks the real code** — the program *is* the spec; no re-encoding of semantics.
- ✓ **Bit-precise IEEE-754, built in** — exactly where fusion breaks (reassociation, rounding); plain reals answer the *wrong* question, and a Lean IEEE-754 library needs manual proofs + gives no counterexamples.
- ✓ **Counterexamples** — a concrete failing tensor when the rewrite is wrong (Lean gives none); e.g. `reassoc_fusion_exact` yields a 1-ULP witness.
- ✓ **Push-button / CI-able** — gate optimisations automatically.

Choose Lean (ITP) only for the *different* goal of a **size-general, all-shapes theorem over reals** — at higher encoding/proof cost, no FP, no counterexamples.

Caveats: ESBMC is **bounded** (verify the shapes you ship) and **OM-dependent**.

Status · what's next

- ✓ QKV fully verified (exact + tolerance, 2 encodings); bias-fused linear added.
- → **Scale**: larger/symbolic dims via k-induction; native `cat`/`split` (currently unwinding-heavy, #5121).
- ✓ **Lean + agent** comparison *done* (see `lean/`): order-preserving fusion is `rfl`/∀-dims; FP reassociation needs BMC.

Scalability is bounded by **SMT solving over floating-point arithmetic** and **loop unwinding** — cost grows rapidly with tensor size and FP-operation count.

Summary

Bounded model checking gives an **automatic, bit-precise, counterexample-producing** equivalence check for PyTorch rewrites — at the cost of being **bounded**.

The encoding cost is **relatively low once the operational model is available** (the program acts as the spec); interactive proof buys generality at a higher encoding/proof cost and **no counterexamples**.

All results are reproducible via the ESBMC harnesses and the `torch` operational model (`make verify`).